

Exercise 0: Prerequisites

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Problem 1 (Operator Norm):Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ be a rectangular matrix with $m \geq n$.

- a) Recall the definition of the operator norm of
- $\mathbf{A}$
- :

$$\|\mathbf{A}\|_2 = \sup_{\mathbf{x} \in \mathbb{R}^n \setminus \{0\}} \frac{\|\mathbf{A}\mathbf{x}\|}{\|\mathbf{x}\|}.$$

Write down the definition of a vector norm. Show that the operator norm is indeed a vector norm in the vector space $\mathbb{R}^{m \times n}$ of matrices.

- b) By the singular value decomposition (SVD) of
- $\mathbf{A}$
- , show that
- $\|\mathbf{A}\|_2 = s_1(\mathbf{A})$
- , the largest singular value of
- $\mathbf{A}$
- .

- c) By SVD of
- $\mathbf{A}$
- again, show that
- $s_1(\mathbf{A}) = \sqrt{\lambda_1(\mathbf{A}^\top \mathbf{A})}$
- , where
- $\lambda_1(\cdot)$
- denotes the largest eigenvalue.

Problem 2 (Calculus):

- a) Let
- $\mathbf{A} \in \mathbb{R}^{m \times n}$
- be a rectangular matrix,
- $\mathbf{x} \in \mathbb{R}^n, \mathbf{y} \in \mathbb{R}^m$
- be column vectors. Let
- $f(\mathbf{x}) = \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2$
- . Compute its gradient
- $\nabla f \in \mathbb{R}^n$
- .

- b) Let
- $\mathbf{U} \in \mathbb{R}^{n \times p}, \mathbf{V} \in \mathbb{R}^{p \times m}$
- , and
- $\mathbf{R} \in \mathbb{R}^{n \times m}$
- . The matrix factorization problem tries to find an approximation of
- $\mathbf{R}$
- as the product of two matrices with smaller common dimension
- p
- , that is

$$\mathbf{R} \approx \mathbf{UV}.$$

This problem can be solved for instance by minimizing the loss $L(\mathbf{U}, \mathbf{V}) := \frac{1}{2} \|\mathbf{UV} - \mathbf{R}\|_F^2$, where $\|\mathbf{A}\|_F := \sqrt{\sum_i \sum_j |A_{ij}|^2}$ is the Frobenius norm.

Compute the derivative of L w.r.t. $\mathbf{U}$, $\frac{\partial L}{\partial \mathbf{U}}$, and w.r.t. $\mathbf{V}$, $\frac{\partial L}{\partial \mathbf{V}}$, respectively.

- c) Consider the problem of non-linear least squares regression with some non-linear function
- $\ell: \mathbb{R} \rightarrow \mathbb{R}$
- and
- n
- data samples
- $(\mathbf{x}_i, y_i)$
- ,

$$L(\mathbf{w}) = \frac{1}{2n} \sum_{i=1}^n (y_i - \ell(\mathbf{x}_i^\top \mathbf{w}))^2.$$

Compute the gradient $\nabla_{\mathbf{w}} L(\mathbf{w})$ and the Hessian $\nabla_{\mathbf{w}}^2 L(\mathbf{w})$.

Problem 3 (Taylor Expansion):Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a twice differentiable function with a local minimum $\mathbf{x}^*$.

- a) Write down the definition of the Hessian $\mathbf{H}(\mathbf{x}) = \nabla^2 f(\mathbf{x}) \in \mathbb{R}^{n \times n}$ of f and the order-2 Taylor expansion of f at $\mathbf{x}^*$.
- b) Using Chain rule or otherwise, prove that for any $\mathbf{x}, \mathbf{v} \in \mathbb{R}^n$, the matrix-vector product $\mathbf{H}(\mathbf{x})\mathbf{v} \in \mathbb{R}^n$ is equal to a limit:

$$\mathbf{H}(\mathbf{x})\mathbf{v} = \lim_{t \rightarrow 0} \frac{\nabla f(\mathbf{x} + t\mathbf{v}) - \nabla f(\mathbf{x})}{t}.$$

- c) Suppose
- f
- is a
- L
- smooth function, that is, there is an
- $L > 0$
- such that

$$\|\nabla f(\mathbf{x}_1) - \nabla f(\mathbf{x}_2)\| \leq L \|\mathbf{x}_1 - \mathbf{x}_2\|, \forall \mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n.$$

Show that each eigenvalue of $\mathbf{H}(\mathbf{x})$ of any $\mathbf{x} \in \mathbb{R}^n$ is upper bounded by L .

- d) Recall that
- $\mathbf{x}^*$
- is a local minimum. Using Problem 1c) and 3c) or otherwise, prove that

$$\|\mathbf{H}(\mathbf{x}^*)\|_2 \leq L$$

e) Using Cauchy-Schwartz inequality and Problem 3d) or otherwise, show that

$$f(\mathbf{x}) - f(\mathbf{x}^*) \leq \frac{L}{2} \|\mathbf{x} - \mathbf{x}^*\|^2 + o\left(\|\mathbf{x} - \mathbf{x}^*\|^3\right).$$

Problem 4 (Expectations, conditioning, and variance):

Let X, Y be discrete random variables with finite second moments.

a) Show that

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

b) Starting from

$$P(X = x) = \sum_y P(X = x \mid Y = y)P(Y = y),$$

prove $\mathbb{E}[X] = \mathbb{E}_Y[\mathbb{E}[X \mid Y]]$.

c) Prove

$$\text{Var}_X(X) = \mathbb{E}_Y[\text{Var}(X \mid Y)] + \text{Var}_Y(\mathbb{E}[X \mid Y]).$$

Here, $\mathbb{E}[X \mid Y]$ and $\text{Var}(X \mid Y)$ should be viewed as functions of the random variable Y .

d) Finally, let $Y \sim \text{Bernoulli}(p)$, and suppose

$$\mathbb{E}[X \mid Y = i] = \mu_i, \quad \text{Var}(X \mid Y = i) = \sigma_i^2, \quad i \in \{0, 1\}.$$

Compute $\mathbb{E}[X]$ and $\text{Var}(X)$. Identify which part of $\text{Var}(X)$ describes variability within the two conditional distributions and which part describes variability between their conditional means. What happens to these two terms if (i) $\mu_0 = \mu_1$, or (ii) $\sigma_0^2 = \sigma_1^2 = 0$?